Hw 1

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Question 1

(a) Particle in a box → ground state energy

From class, we know TISE solution for two particles in PIB is :

$$\phi_1 = \sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_1}{L}\right)$$

$$\phi_2 = \sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_2}{L}\right)$$

; Shown from separation of variables & applying BC's where $\psi_{n_1,n_2} = \phi_1(x)\,\phi_2(x)$

Since we are dealing w/ fermions (w/ half-integer spins), we can use the slater determinant to obtain the new wave function:

$$\psi = \frac{1}{\sqrt{2}}\begin{vmatrix}\phi_1(x_1) & \phi_2(x_1)\\ \phi_1(x_2) & \phi_2(x_2)\end{vmatrix}$$

$$\psi(x_1,x_2) = \frac{1}{\sqrt{2}}\left[\phi_1(x_1)\,\phi_2(x_2) - \phi_1(x_2)\,\phi_2(x_1)\right]$$

$$\boxed{\psi(x_1,x_2) = \frac{1}{\sqrt{2}}\left[\sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_1}{L}\right)\cdot\sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_2}{L}\right) - \sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_2}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_1}{L}\right)\right]}$$

(b) Expectation energy of the system : $E_{GS} = \dfrac{\iint \psi^* H \psi \, dx_1\, dx_2}{\iint \psi^*\psi \, dx_1\, dx_2}$

Using the $\psi(x_1,x_2)$ expression from part (a), we can solve for E_{GS} :

$$\hat{H} = \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_i^2} \quad ; \text{ for 2 particles: } \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x_1^2} - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_2^2} = \hat{H}$$

First derivative: $\dfrac{\partial \psi}{\partial x_1} = \dfrac{1}{\sqrt{2}}\left[+\sqrt{\dfrac{2}{L}}\cos\left(\dfrac{n_1\pi x_1}{L}\right)\cdot\dfrac{n_1\pi}{L}\cdot\sqrt{\dfrac{2}{L}}\sin\left(\dfrac{n_2\pi\cdot x_2}{L}\right) - \dfrac{2}{L}\cos\left(\dfrac{n_2\pi x_1}{L}\right)\dfrac{n_2\pi}{L}\cdot\sin\left(\dfrac{n_1\pi x_2}{L}\right)\right]$

$$= \frac{\sqrt{2}}{L}\cdot\frac{n_1\pi}{L}\cos\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right) - \frac{\sqrt{2}}{L}\frac{n_2\pi}{L}\cos\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)$$

$$\frac{\partial^2\psi}{\partial x_1^2} = \frac{-\sqrt{2}}{L}\cdot\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right) + \frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)$$

Similarly,

$$\frac{\partial^2\psi}{\partial x_2^2} = \frac{-\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right) + \frac{\sqrt{2}}{L}\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)$$

Since ψ has no complex parts, $\psi^* = \psi$

$$\therefore \quad \psi^* = \frac{1}{\sqrt{2}}\left[\overbrace{\frac{2}{L}\cdot\sin\left(\frac{n_1\pi x_1}{L}\right)\cdot\sin\left(\frac{n_2\pi x_2}{L}\right)}^{A_1} - \overbrace{\frac{2}{L}\cdot\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)}^{A_2}\right]$$

$$\hat{H}\psi = \frac{-\hbar^2}{2m}\left[\frac{\partial^2\psi}{\partial x_1^2} + \frac{\partial^2\psi}{\partial x_2^2}\right] = \frac{-\hbar^2}{2m}\left[\underbrace{-\frac{\sqrt{2}}{L}\cdot\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)}_{B_1} + \underbrace{\frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)}_{B_2} \underbrace{-\frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)}_{B_3} + \underbrace{\frac{\sqrt{2}}{L}\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)}_{B_4}\right]$$

Normalization: $\displaystyle\int_0^L\!\!\int_0^L \psi^*\psi \, dx_1\, dx_2 = \int_0^L\!\!\int_0^L \frac{1}{2}\cdot\left[\frac{2^2}{L^2}\sin^2\left(\frac{n_1\pi x_1}{L}\right)\cdot\sin^2\left(\frac{n_2\pi x_1}{L}\right) + \frac{2^2}{L^2}\sin^2\left(\frac{n_1\pi x_2}{L}\right)\sin^2\left(\frac{n_2\pi x_1}{L}\right) - \frac{8}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\right]\cdot dx_1\, dx_2$

$$= \frac{1}{2}\left[\int_0^L \frac{4}{L^2}\sin^2\left(\underbrace{\frac{n\pi x_1}{L}}_{a}\right)\sin^2\left(\underbrace{\frac{n_2\pi x_1}{L}}_{b}\right)\cdot dx_1\cdot\int_0^L 1\cdot dx_2\right] + \frac{1}{2}\left[\int_0^L \frac{4}{L^2}\sin^2\frac{n_1\pi x_2}{L}\cdot dx_2\cdot\int_0^L \sin^2\frac{n_2\pi x_1}{L}\cdot dx_1\right] - \frac{1}{2}\left[\int_0^L \frac{8}{L^2}\cancelto{0}{\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)}\cdot dx_1\int_0^L \cancelto{0}{\sin\left(\frac{n_2\pi x_2}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)}\cdot dx_2\right]$$

$$= \frac{1}{2}\,\frac{\cancel{4}}{\cancel{L^2}}\cdot\frac{\cancel{L^2}}{\cancel{4}} + \frac{1}{2}\cdot\frac{\cancel{4}}{\cancel{L^2}}\cdot\frac{\cancel{L^2}}{\cancel{4}}$$

$$= 1 \ ✓$$

$$\int_0^L\!\!\int_0^L \psi^*\hat{H}\psi\cdot dx_1\, dx_2 = \int_0^L\!\!\int_0^L \frac{-\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}}\cdot\frac{\hbar^2}{2m\cdot L^2}\left[-A_1\cdot B_1 + A_1\cdot B_2 - A_1\cdot B_3 + A_1\cdot B_4 + A_2\cdot B_1 - A_2\cdot B_2 + A_2\cdot B_3 - A_2\cdot B_4\right]\cdot dx_1\, dx_2$$

Solving for individual terms :

$$\underbrace{\left(\frac{n_1^2\pi^2}{L^2}\right)}_{C_1}\iint -A_1\cdot B_1 = C_1\int_0^L\!\!\int \sin^2\left(\frac{n_1\pi x_1}{L}\right)\sin^2\left(\frac{n_2\pi x_2}{L}\right)\cdot dx_1\, dx_2 \quad ;$$

& $\sin(n\pi) = 0$ for $n = 0, \pm1, \pm2$

$$= C_1\int_0^L \sin^2\left(\frac{n_1\pi x_1}{L}\right)\cdot dx_1 \int_0^L \sin^2\left(\frac{n_2\pi x_2}{L}\right)\cdot dx_2$$

$$= C_1\left(\frac{L}{2}\cdot\frac{L}{2}\right)$$

$$= \frac{n_1^2\pi^2}{\cancel{L^2}}\cdot\frac{\cancel{L^2}}{4} = -\frac{n_1^2\pi^2}{4}$$

$$\left(\frac{n_2^2\pi^2}{L^2}\right)\iint A_1\cdot B_2 = \iint_0^L \sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\cdot dx_1\,dx_2 \Rightarrow \int_0^L \sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\cdot dx_1 \rightarrow$$

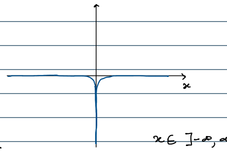
Since sin is an even function
↓
This integral evaluated at 0 to L for interger n's will evaluate to 0

Similarly solving for individual terms we only get $-A_1B_1$, $-A_1B_3$, $-A_2B_2$ & $-A_2B_4$ to survive: → each corresponding to n_1, n_2 energy level

That yields us w/ the desired answer: $\boxed{E_{GS} = \frac{-\hbar^2}{2mL^2}\left[\frac{-2(n_1\pi)^2}{4} - \frac{2(n_2\pi)^2}{4}\right] = \varepsilon_1 + \varepsilon_2 \quad ; \quad \varepsilon_i = \frac{n_i^2\pi^2\hbar^2}{2mL^2}}$

Question 2

δ function @ $V_{ext}(x) = -\alpha\,\delta(x)$



BC's: $\Psi \rightarrow 0$ @ $x \rightarrow \pm\infty$

$\Psi(x) = \frac{\sqrt{m\alpha}}{\hbar}\exp\left(-\frac{m\alpha}{\hbar^2}|x|\right)$; $E = -\frac{m\alpha^2}{2\hbar^2}$

$x \in]-\infty, \infty$

↳ These are given solutions to the full TISE

Using a DFT approach:

$$E[\rho] = \int_{-\infty}^{\infty}\frac{\pi^2\hbar^2}{6m}\rho^3\cdot dx - \alpha\int_{-\infty}^{\infty}\delta(x)\rho\cdot dx \quad \rightarrow$$

Our energy functional

Given constrains: $\int_{-\infty}^{\infty}\rho(x)\cdot dx = 1$ (Assuming dealing w/ N particles)

↳ Constructing the lagrangian functional w/ the given constraint:

$$F[\rho] = E[\rho] - \mu\left(\int_{-\infty}^{\infty}\rho\cdot dx - N\right) = \int f(\rho(x))\cdot dx - \alpha\rho(0) + \mu N$$

We know for functionals:

$$\frac{\delta F[\rho(x)]}{\delta\rho(x)} = \frac{\partial f}{\partial\rho} - \nabla\cdot\frac{\partial f}{\partial\nabla_\rho}$$; where f is the integrand as shown above

Since f does not have any derivatives of f: $\frac{\delta F}{\delta\rho} = \frac{\partial f}{\partial\rho}$

$$\frac{\delta F}{\delta\rho} = 0 = \frac{\partial}{\partial\rho}\left(\frac{\pi^2\hbar^2}{6m}\rho^3 - \mu\rho\right) - \cancelto{0}{\frac{\partial}{\partial\rho}(\alpha\rho(0))} + \cancelto{0}{\frac{\partial}{\partial\rho}(\mu N)}$$

$$0 = \frac{\pi^2\hbar^2}{2m}\rho^2 - \mu$$

$$\rho_{min} = \sqrt{\frac{2m\mu}{\pi^2\hbar^2}}$$

Plugging into constraint:

$$\int_{-\infty}^{\infty}\sqrt{\frac{2m\mu}{\pi^2\hbar^2}}\cdot dx = N$$

$$\frac{2m\mu}{\pi^2\hbar^2}\underbrace{\int_{-\infty}^{\infty}\cdot dx}_{\text{This integral diverges and cannot be evaluated}} = N$$

This integral does not converge, therefore cannot find an expression for ρ_{min} via DFT approach.

This issue could be due to the unbounded box sides which means that the e^- can be spread across the infinite distance, which is an uphysical system.

Question 3

$V_{ext} = \frac{1}{2}kx^2$; $x \in]-\infty, \infty[$

(a) Energy functional:

$$E[\rho] = \int_{-\infty}^{\infty}\frac{\pi^2\hbar^2}{6m}\rho^3\,dx + \int_{-\infty}^{\infty}\frac{1}{2}kx^2\rho(x)\cdot dx$$

(b) Constraint: $\int_{-\infty}^{\infty}\rho(x)\cdot dx = N$ (Normalization)

$$\therefore F[\rho] = E[\rho] - \mu\left(\int_{-\infty}^{\infty}\rho(x)\cdot dx - N\right) = \int\overbrace{\left(\frac{\pi^2\hbar^2}{6m}\rho^3\cdot dx + \frac{1}{2}kx^2\rho - \mu\rho\right)}^{f}\cdot dx + \mu N$$

Minimizing functional $\frac{\delta F}{\delta\rho} = 0$;

$$\frac{\delta F}{\delta\rho} = \frac{\partial f}{\partial\rho} - \cancelto{0}{\nabla\cdot\frac{\partial f}{\partial\nabla_\rho}} = \frac{\partial}{\partial\rho}\left(\frac{\pi^2\hbar^2}{6m}\rho^3 + \frac{1}{2}kx^2\rho - \mu\rho\right) + \cancelto{0}{\frac{\partial}{\partial\rho}(\mu N)}$$

$$0 = \frac{\pi^2\hbar^2}{2m}\rho^2 + \frac{1}{2}kx^2 - \mu$$

$$\frac{\delta F}{\delta \rho} = \frac{\partial f}{\partial \rho} - \cancel{\nabla \cdot \frac{\partial f}{\partial \nabla \rho}}^{\,0} = \frac{\partial}{\partial \rho}\left(\frac{\pi^2\hbar^2}{6m}\rho^3 + \frac{1}{2}kx^2\rho - \mu\rho\right) + \cancel{\frac{\partial}{\partial \rho}(\mu N)}$$

$$0 = \frac{\pi^2\hbar^2}{2m}\rho^2 + \frac{1}{2}kx^2 - \mu$$

$$\rho_{min} = \left[\left(\mu - \frac{1}{2}kx^2\right)\frac{2m}{\pi^2\hbar^2}\right]^{1/2}$$

$$\boxed{\rho_{min} = \frac{\sqrt{2m}}{\pi\hbar}\left(\mu - \frac{1}{2}kx^2\right)^{1/2}} \rightarrow$$

For physical solutions

$$\mu - \frac{1}{2}kx^2 > 0$$

$$\mu > \frac{1}{2}kx^2$$

Solving for μ:

$$\int_{-\infty}^{\infty} \rho_{min}(x) \cdot dx = N$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\sqrt{2m}}{\pi\hbar}\left(\mu - \frac{1}{2}kx^2\right)^{1/2} \cdot dx = N$$

$$\Rightarrow \int_{-\infty}^{\infty}\left(\mu - \frac{1}{2}kx^2\right)^{1/2} \cdot dx = \frac{N\pi\hbar}{\sqrt{2m}} \quad ; \quad \text{Since } \mu > \frac{1}{2}kx^2$$

$$x^2 > \frac{2\mu}{k}$$

$$-\sqrt{\frac{2\mu}{k}} < x < \sqrt{\frac{2\mu}{k}}$$

The Integration limits: change to

$$\int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \left(\mu - \frac{1}{2}kx^2\right)^{1/2} = \frac{N\pi\hbar}{\sqrt{2m}}$$

Using Wolfram Alpha:

$$\left[\frac{\mu}{2\sqrt{\frac{1}{2}k}} \cdot \arcsin\left(\frac{\sqrt{\frac{1}{2}k}\,x}{\sqrt{\mu}}\right) + \frac{x}{2}\sqrt{\mu - \frac{1}{2}kx^2}\right]_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} = \frac{N\pi\hbar}{\sqrt{2m}}$$

$$\Rightarrow \frac{\pi\mu}{2\sqrt{\frac{1}{2}k}} = \frac{N\pi\hbar}{\sqrt{2m}}$$

$$\therefore \boxed{\mu = \frac{N\hbar\sqrt{k}}{\sqrt{m}}}$$

Plugging back into ρ_{min}: $$\rho_{min} = \left[\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}} - \frac{1}{2}kx^2\right)\frac{2m}{\pi^2\hbar^2}\right]^{1/2}$$

(c) $$E[\rho_{min}] = \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \frac{\pi^2\hbar^2}{6m}\rho_{min}^3\,dx + \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \frac{1}{2}kx^2\rho_{min}(x) \cdot dx$$; The bounds are changed to exclude unphysical regions ; $\mu = \frac{N\hbar\sqrt{k}}{\sqrt{m}}$

$$= \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2} \cdot \frac{\pi^2\hbar^2}{6m} \cdot \left(\frac{N\hbar\sqrt{k}}{\sqrt{m}} - \frac{1}{2}kx^2\right)^{3/2} + \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \frac{1}{2}kx^2 \cdot \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2} \cdot \left(\frac{N\hbar\sqrt{k}}{\sqrt{m}} - \frac{1}{2}kx^2\right)^{1/2} \cdot dx$$

Using Wolfram Alpha

$$= \frac{\sqrt{2}}{\cancel{3}}\frac{1}{\cancel{\pi}\hbar}\sqrt{m} \cdot \frac{\cancel{3\pi}}{8\sqrt{\frac{1}{2}k}} \cdot \left(\frac{N\hbar\sqrt{k}}{\sqrt{m}}\right)^2 + \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2} \cdot \frac{\pi\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}}\right)^2}{8\sqrt{\frac{1}{2}k}}$$

$$= \frac{1}{4}\sqrt{\frac{k}{m}} \cdot \frac{N^2\hbar k}{m} + \frac{\cancel{m}}{\cancel{\pi\hbar}} \cdot \frac{\cancel{\pi}N^2\hbar^{\cancel{2}}k}{4\cancel{m}\sqrt{k}}$$

$$\left(\frac{2m}{\cancel{\pi}^2\hbar^2}\right)^{3/2} \cdot \frac{\cancel{\pi}^2\hbar^2}{\underset{2}{\cancel{6}}m} \frac{\cancel{3}\pi a^2}{8\sqrt{b}} + \frac{\pi a^2}{8\sqrt{b}}\sqrt{\frac{2m}{\pi^2\hbar^2}}$$

$$\frac{\pi a^2}{8\sqrt{b}}\left(\frac{\sqrt{2m}}{\hbar\pi} + \frac{\sqrt{2m}}{\pi\hbar}\right)$$

$$= \frac{\cancel{\pi}a^2}{\underset{4}{\cancel{8}}\sqrt{b}} \cdot \frac{\cancel{2}\sqrt{2m}}{\cancel{\pi}\hbar}$$

$$= N^2\hbar^{\cancel{2}}\frac{k}{m} \cdot \frac{\cancel{4}}{\cancel{8}}\frac{1}{4} \cdot \frac{\sqrt{2}}{\sqrt{k}} \cdot \frac{\sqrt{2m}}{\cancel{\hbar}}$$

$$= \frac{1}{2}N^2\hbar\frac{\sqrt{k}}{\sqrt{m}}$$

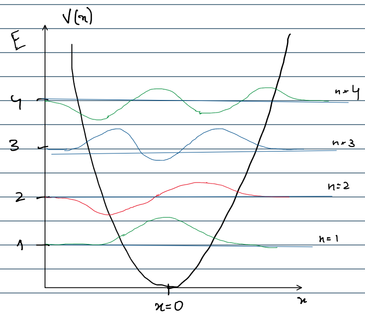
$$\boxed{E[\rho_{min}] = \frac{N^2\hbar}{2}\frac{\sqrt{k}}{\sqrt{m}}}$$

(d) Harmonic Oscillator : $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$; $n = 0, 1, 2, 3 \ldots$; $\omega = \sqrt{\frac{k}{m}}$

$$\psi_n(x) = N_n e^{-\beta^2 x^2/2} H_n(\beta x) \quad ; \quad n = 0, 1, 2, 3$$

where $\beta = \sqrt{\frac{m\omega}{\hbar}}$

$H_n(y) \rightarrow$ hermite polynomial



N	DFT	Harmonic
1	$\frac{1}{2}\hbar\sqrt{\frac{k}{m}}$	$\frac{3}{2}\hbar\sqrt{\frac{k}{m}}$
2	$2\hbar\sqrt{\frac{k}{m}}$	$\frac{5}{2}\hbar\sqrt{\frac{k}{m}}$
3	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$	$\frac{7}{2}\hbar\sqrt{\frac{k}{m}}$
4	$8\hbar\sqrt{\frac{k}{m}}$	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$

DFT overapproximates the energy

3	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$	$\frac{7}{2}\hbar\sqrt{\frac{k}{m}}$
4	$8\hbar\sqrt{\frac{k}{m}}$	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$
5	$\frac{25\hbar}{2}\sqrt{\frac{k}{m}}$	$\frac{11}{2}\hbar\sqrt{\frac{k}{m}}$

Question 4

Non-interacting & spherical symmetry

(a) Doing similar analysis:

$$F[\rho] = E[\rho] - \mu\left(\int \rho \cdot d\vec{r}^{3} - N\right) = \int\left(\rho^{5/3}\cdot\kappa - \gamma\cdot\frac{\rho}{r} - \mu\rho\right)4\pi r^2\cdot dr + \mu N$$

$$\frac{\delta F}{\delta\rho} = 0 = \frac{\partial}{\partial\rho}\left(\left(\rho^{5/3}\cdot\kappa - \frac{\gamma\rho}{r} - \mu\rho\right)4\pi r^2\right) + \cancel{\frac{\partial}{\partial\rho}(\mu N)}\xrightarrow{} 0$$

$$0 = 4\pi\left(\kappa\frac{5}{3}\rho^{2/3}r^2 - \gamma r - \mu r^2\right)$$

$$\frac{5}{3}\kappa r^2\cdot\rho^{2/3} = \gamma r + \mu r^2$$

$$\rho^{2/3} = \frac{3}{5}\left(\frac{\gamma}{r}+\mu\right)\frac{1}{\kappa}$$

$$\rho_{min} = \left(\frac{3}{5\kappa}\left(\frac{\gamma}{r}+\mu\right)\right)^{3/2}$$

Finding μ: $\int_0^{\frac{\gamma}{|\mu|}} \rho_{min}\cdot d\vec{r}^{3} = N$; For physical solutions, $\left(\mu+\frac{\gamma}{r}\right) > 0$

$\frac{\gamma}{r} > -\mu$

$r > -\frac{\gamma}{\mu}$ ↗

Since r cannot be negative and γ has to also be positive,

$\mu < 0$; $\mu = -|\mu|$

s.t. $r < \frac{\gamma}{|\mu|}$

$$\int_0^{\frac{\gamma}{|\mu|}}\left(\frac{3}{5\kappa}\right)^{3/2}\cdot\left(\frac{\gamma}{r}-|\mu|\right)^{3/2}\cdot 4\pi r^2\cdot dr = N$$

$$\int_0^{\frac{\gamma}{|\mu|}}\left(\frac{\gamma}{r}-|\mu|\right)^{3/2} r^2\cdot dr = \frac{N}{4\pi}\cdot\left(\frac{5\kappa}{3}\right)^{3/2}$$

Using wolfram Alpha:

$$\frac{\pi\gamma^3}{{}_{4}\,\cancel{16}(|\mu|)^{3/2}} = \frac{N}{\cancel{4}\pi}\left(\frac{5\kappa}{3}\right)^{3/2}$$

$$(|\mu|)^{3/2} = \frac{1}{4}\frac{\pi^2\gamma^3}{N}\cdot\left(\frac{3}{5\kappa}\right)^{3/2}$$

$$|\mu| = \left(\frac{1}{4}\frac{\pi^2}{N}\right)^{2/3}\cdot\gamma^2\cdot\frac{3}{5\kappa}$$

$$|\mu| = \frac{\gamma^2 3}{5\kappa}\left(\frac{\pi^2}{4N}\right)^{2/3}$$

Since $\mu = -|\mu|$;

$$\boxed{\mu = -\frac{\gamma^2\cdot 3}{5\kappa}\left(\frac{\pi^2}{4N}\right)^{2/3}}$$

(b) $E[\rho_{min}] = ?$

Plugging in μ in ρ_{min} expression: $\rho_{min} = \left[\frac{3}{5\kappa}\frac{\gamma}{r} - \frac{3}{5\kappa}|\mu|\right]^{3/2}$

$$\rho_{min} = \left(\frac{3}{5\kappa}\left(\frac{\gamma}{r} - \frac{\gamma^2\cdot 3}{5\kappa}\left(\frac{\pi^2}{4N}\right)^{2/3}\right)\right)^{3/2}$$

$$\rho_{min} = \left[\frac{3\gamma}{5\kappa\cdot r} - \frac{9\gamma^2}{25\kappa^2}\left(\frac{\pi^2}{4N}\right)^{2/3}\right]^{3/2}$$

$$E[\rho_{min}] = \kappa\int_0^{\frac{\gamma}{|\mu|}}\left(\frac{3}{5\kappa}\frac{\gamma}{r} - \frac{3}{5\kappa}|\mu|\right)^{5/2}4\pi r^2\cdot dr - \gamma\int_0^{\gamma/|\mu|}\left(\frac{3}{5\kappa}\frac{\gamma}{r} - \frac{3}{5\kappa}|\mu|\right)^{3/2}\frac{1}{r}\cdot 4\pi r^2\cdot dr$$

$$= \left(\frac{3}{5\kappa}\right)^{5/2}\cdot\kappa\cdot 4\pi\int_0^{\gamma/|\mu|}\left(\frac{\gamma}{r}-|\mu|\right)^{5/2}r^2\cdot dr - \gamma\left(\frac{3}{5\kappa}\right)^{3/2}4\pi\int_0^{\gamma/|\mu|}\left(\frac{\gamma}{r}-|\mu|\right)^{3/2}\cdot r\cdot dr$$

$$= \left(\frac{3}{5\kappa}\right)^{5/2}\cdot\kappa\cdot 4\pi\cdot\frac{5\pi\gamma^3}{16\sqrt{|\mu|}} - 4\pi\gamma\left(\frac{3}{5\kappa}\right)^{3/2}\cdot\frac{3\pi\gamma^2}{8\sqrt{|\mu|}}$$

$$= \frac{4\pi\gamma^3}{\sqrt{|\mu|}}\left(\frac{3}{5\kappa}\right)^{3/2}\left(\frac{3}{\cancel{5\kappa}}\cdot\cancel{\kappa}\cdot\frac{\cancel{5}\pi}{16} - \frac{3\pi}{8}\right)$$

$$\left(5\kappa\right) \quad \frac{}{16\sqrt{|\mu|}} \quad \left(\frac{}{5\kappa}\right) \quad \frac{}{8\sqrt{|\mu|}}$$

$$= \frac{4\pi\gamma^3}{\sqrt{|\mu|}}\left(\frac{3}{5\kappa}\right)^{3/2}\left(\frac{3}{\cancel{5\kappa}}\cdot\cancel{\kappa}\cdot\frac{\cancel{5}\pi}{16} - \frac{3\pi}{8}\right)$$

$$= \frac{\cancel{4}\pi\gamma^3}{\sqrt{|\mu|}}\left(\frac{3}{5\kappa}\right)^{3/2}\left(-\frac{\cancel{2}\pi}{\cancel{16}}\right) \quad \cancel{8}\ 2$$

$$\frac{1\times2}{3\times2} - \frac{3\times3}{2\times3} = -\frac{7}{6}$$

$$E[\rho_{min}] = -\frac{\pi^2\gamma^3}{2}\left(\frac{3}{5\kappa}\right)^{3/2}\cdot\left(\frac{5\kappa}{3\gamma^2}\right)^{1/2}\left(\frac{\pi^2}{4N}\right)^{1/3}$$

$$\frac{(5\kappa)^{1/2}}{(5\kappa)^{3/2}} = 5\kappa^{-2/2}$$

$$\frac{3}{1} - \frac{2}{3} = \frac{7}{3}$$

$$2 + \frac{2}{3}$$

$$\boxed{E[\rho_{min}] = -\frac{3}{10\kappa}\gamma^2\pi^2\cdot\left(\frac{\pi^2}{4N}\right)^{1/3}}$$

$$E[\rho_{min}] \propto \left(\frac{1}{N}\right)^{1/3}$$

(c) $2e^-$ per orbit

Solution to TISE for H_2 atom: ; $n > l \geq |m|$

$$\psi_{nlm} = R_{nl}(r)\,Y_l^m(\theta,\phi)$$

$$E_n = \frac{-\hbar^2}{2m_e a_0^2}\cdot\frac{1}{n^2} \ ; \ a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} \ ; \ E_1 = -13.61\text{ eV}$$

$\psi_{100} \rightarrow$ corresponds to ground state.

$E_n \propto \frac{1}{n^2}$ while our prediction has $E_n \propto \frac{1}{N^3}$

$\psi_{210} \rightarrow n=2$ w/ azimuthal no. or p-states

e^- will fill up n, then l & the m